The Oracle Gap Has a Floor: Separating Aleatoric and Deployable Headroom in Evaluating Label Acquisition

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Abstract. Oracle selectors in label-acquisition benchmarks can exploit both realized training outcomes and the evaluation set. We distinguish the value of perfect information about hidden training labels from test-set reuse and from the performance of a specified ex-ante selector. For a fixed feasible family, we derive distribution-free population inequalities and their exact empirical analogues; a Gaussian two-set expression is an approximation rather than a universal bound. Across 624 simulated states, the empirical inequalities hold throughout. Eight cohorts and three downstream learners reveal substantial variation in hindsight opportunity. Within-cohort label-flip interventions increase the empirical family floor monotonically in all three tested cohorts, while larger candidate pools show heterogeneous effects. Exact enumeration on 12-candidate pools checks a greedy-oracle approximation, and probability-model perturbations expose sensitivity beyond held-out calibration checks. We retain cross-fitted greedy performance as an operational benchmark, with its optimization and Monte Carlo errors stated explicitly. Delayed replay shows that DC-PRISM can improve on quantity-oriented references while losing to information-greedy rules. The resulting protocol audits which parts of an oracle gap are supported by an evaluation design, rather than treating the entire gap as attainable improvement.

Keywords: Active learning · Evaluation · Oracle bounds · Aleatoric uncertainty · Risk-constrained acquisition

1 Introduction

How much better could an acquisition strategy be? Active-learning research increasingly answers with an *oracle*: a selector that, unlike any deployable strategy, may look at the labels it acquires and at the evaluation set, and so traces the best attainable learning curve. Recent benchmarks construct greedy oracles that add, step by step, the candidate whose *realized* label most improves test performance [42,43], and best-of-strategies oracles that pick, per batch, the strategy whose labelled batch helps most [18]. The gap between oracle and practical strategies is then read as headroom: state-of-the-art strategies are reported to "fall noticeably short of oracle performance, especially in large-scale

datasets" [18]. This complements a broader effort to make active-learning evaluation reliable [28,19].

An oracle that selects with realized labels is an *ex-post* benchmark. Part of its advantage is unreachable by construction: when outcomes are random given the covariates, some feasible label set will turn out to be lucky, and only hindsight can find it. The distinction between ex-post and ex-ante regret is familiar in decision-focused learning and robust optimization [34], and in bandits the gap between expected regret and pseudo-regret grows with the noise of the arms [6,2]. We investigate its magnitude for oracle bounds in label acquisition, where it affects whether an "oracle gap" signals room for better strategies or merely outcome noise. Labels in credit, healthcare and marketing record stochastic outcomes rather than near-deterministic categories, so the question matters most where labels are bought with consequential actions.

Contributions.

- A protocol distinguishing fixed-family perfect-information value, evaluation-set reuse, and the gain of a specified cross-fitted selector. These quantities have different identification and estimation requirements.
- Distribution-free inequalities for the fixed-family floor, checked on 624 states, together with exact small-pool enumeration, probability-model sensitivity, and within-cohort label-noise interventions.
- An empirical audit across eight cohorts and three learners, with delayed replay as a separate outcome check. The evidence documents where hindsight opportunity persists and where practical selection rules fail to exploit it.

2 Setting

At round t a learner holds labeled history D_t, the covariates of pending acquisitions O_t (selected but not yet resolved), an unlabeled candidate pool C_t and target covariates Q_t. Candidate i has a visible exposure $e_i \geq 0$ and a binary outcome Y_i revealed only after acquisition. A strategy chooses $S_t \subseteq C_t$ from visible information $\mathcal{H}_t$. Learning is measured by the Brier score [5] of a downstream learner (an ℓ_2-logistic model, gradient boosting, or a one-hidden-layer network); $L(S; Y)$ is its test loss after fitting on $D \cup O \cup S$, where Y collects the hidden training outcomes of O and S.

Risk budget. Acquiring a label is an action with exposure, as when a lender approves an applicant to observe repayment [25,41]. With a risk score q_i, $\ell = \log(1/\alpha)$ and $\phi(q) = q(1-q)$ for $q \leq \frac{1}{2}$, $\frac{1}{4}$ otherwise, feasible sets satisfy

$$R_\alpha(S) = \sum_{i \in S} e_i q_i + \left(2\ell \sum_{i \in S} e_i^2 \phi(q_i)\right)^{1/2} + \frac{\ell}{3} \max_{i \in S} e_i \leq B_t, \tag{1}$$

a sub-gamma bound [4] under which realized adverse exposure exceeds $R_\alpha(S)$ with probability at most $\alpha = 0.1$ under conditional independence and $p_i \leq q_i$.

Our scores q_i (a Wilson-shifted [44] high quantile of 40 Laplace draws of the fitted logit) do not guarantee $p_i \le q_i$, so (1) is the operational feasible region shared by all strategies. The budget B_t is 1.03 times the envelope of the k lowest-risk candidates ($k = 6$ unless stated); the *quantity reference* M_K is the largest feasible set, found by enumeration for at most 16 candidates and otherwise by an exposure-stratified search that certified its cardinality in 1,320 of 1,408 states ; exact follow-up search certified all remaining 88 reference counts. For benchmark cohorts without a natural exposure we set $e_i = 1$.

3 Decomposing the Oracle Gap

Family oracle. Fix $\mathcal{H}$ and a family $\mathcal{F}$ of feasible sets built from visible information and containing $M = M_K$. With $\bar{L}$ the test loss with test-label noise integrated out,

$$H_{\text{post}} = \mathbb{E}\big[\bar{L}(M;Y) - \min_{S\in\mathcal{F}} \bar{L}(S;Y) \mid \mathcal{H}\big], \tag{2}$$

$$H_{\text{pre}} = \mathbb{E}[\bar{L}(M;Y) \mid \mathcal{H}] - \min_{S\in\mathcal{F}} \mathbb{E}[\bar{L}(S;Y) \mid \mathcal{H}]. \tag{3}$$

H_{post} is the lead of an oracle that picks the best set after seeing the hidden training labels; H_{pre} is the lead of the best fixed choice. Their difference, the *aleatoric floor* $F = H_{\text{post}} - H_{\text{pre}} \ge 0$ (since $\mathbb{E}\min \le \min\mathbb{E}$), is unreachable by any strategy restricted to $\mathcal{F}$; it is the expected value of perfect information about the training outcomes [16]. For a within-family strategy with expected gain Γ_π over M, $H_{\text{post}} - \Gamma_\pi = F + (H_{\text{pre}} - \Gamma_\pi)$, and only the second term, the *selection gap*, can be closed.

Greedy benchmark diagnostics. Benchmarks select a batch greedily using realized labels and an evaluation set [42,43]. We split test records into T_1, T_2. Let G_{rep} be the greedy selector's mean gain when selecting and scoring on T_1, G_{hold} its gain when scoring on T_2, and G_{cf} the gain of a greedy selector choosing by mean T_1 loss on one bank of training-label replicates and evaluated on T_2 using an independent bank, with banks exchanged and averaged. Then

$$G_{\text{rep}} = (G_{\text{rep}} - G_{\text{hold}}) + (G_{\text{hold}} - G_{\text{cf}}) + G_{\text{cf}}.$$

This is an accounting identity. The first contrast diagnoses evaluation-set reuse; the second is a *greedy residual*, which also contains search and finite-simulation effects. It equals the fixed-family floor only when the respective selectors optimize a common family and their objectives and evaluation distributions agree. We therefore do not identify this residual as an irreducible floor or call G_{cf} the optimal deployable value.

Valid fixed-family inequalities. Write $d_S = \bar{L}(M;Y) - \bar{L}(S;Y)$, $\mu_S = \mathbb{E}d_S$, and let b maximize μ_S; let r be another set. The family floor is $F = \mathbb{E}\max_S d_S - \max_S \mu_S$. Pointwise comparison with b, r, and centering each gain, give

$$\mathbb{E}(d_r - d_b)_+ \ \le F \ \le \mathbb{E}\max_S (d_S - \mu_S). \tag{4}$$

Indeed, $\max_S d_S - d_b \geq (d_r - d_b)_+$, whereas $\max_S d_S \leq \max_S \mu_S + \max_S(d_S - \mu_S)$. Replacing expectations and means by the same empirical distribution gives exact sample inequalities, including when b, r are selected from that sample. These empirical statements do not provide confidence coverage for population F. If the centered gains are jointly Gaussian, arbitrary dependence is allowed and $F \leq \max_S \sigma_S \sqrt{2 \log |\mathcal{F}|}$. If $d_b - d_r$ is normal with mean Δ and standard deviation s, the lower expression is $s\varphi(\Delta/s) - \Delta\Phi(-\Delta/s)$. Normality is an additional modeling assumption. An independent-normal construction is not a universal upper bound.

Estimation and model substitution. With known p, common random numbers generate training-label replicates. For a fixed family, maximizing estimated mean gain gives an upward-biased estimate of H_{pre} in expectation. Selecting on one label bank and evaluating on another yields the expected value of that finite-bank selector, at most H_{pre}. Neither expectation statement is a deterministic finite-sample bracket. On real data, a boosting model fitted on disjoint auxiliary records supplies $\hat{p}$; recorded test labels remain fixed. This evaluates opportunity conditional on that test set and probability model. Comparing simulated and recorded-outcome hindsight gaps tests one functional of the model, not full identification of the floor. Taking the per-state maximum of independently noisy policy-value estimates can itself introduce selection optimism; such post-hoc maxima are sensitivity summaries, not certified lower bounds on achievable value.

4 Strategies under Evaluation

All strategies share the state, q, the budget and (1). *Safe-MinRisk* returns M_K; *fixed count* takes a fixed fraction of K by point value; *Fisher-cost* packs by single-label variance reduction per exposure. Four rules pack greedily until no feasible candidate remains: *Random*, *Uncertainty* [27], *BAIT-greedy* on $\operatorname{tr}(F_Q(H_{D+O} + A_S)^{-1})$ [1], and *TargetVar-greedy* on the target-weighted proxy

$$V_\theta(S) = \operatorname{tr}\left[W_Q\{H_{D+O}^{-1} - (H_{D+O} + A_S)^{-1}\}\right], \tag{5}$$

where H_{D+O} is the logistic Fisher information of history and pending covariates, A_S that of S, and W_Q weights target directions by their Brier sensitivity [47,3]. *Proxy-argmax* picks $\arg\max V_{\hat{\theta}}$ over the evaluation family. *DC-PRISM* is a conservative policy designed for risk-constrained acquisition with pending outcomes: it evaluates (5) on a family anchored at M_K under 20 Laplace parameter draws [29], scores each set by the 0.2-quantile of $\tanh\{(V_m(S) - V_m(M_K))/2.5V_m(M_K)\}$ minus a quantity penalty $0.22(K - |S|)_+/K$, and returns M_K unless the best score exceeds 0.08. A selection takes about 50 ms with 24 candidates. All information proxies here are derived from the logistic model.

5 Validating the Protocol

Studies. Every study fixed its states, seeds, comparison families and a material-effect threshold of 5×10^{-4} Brier before running, with protocol and code frozen in version control; we report paired means with 95% Bonferroni simultaneous intervals and Holm-adjusted tests [15]. All values are in 10^{-3} Brier. Cohorts: four credit cohorts (HMEQ, LendingClub, PPDai, UCI Credit [32,26,12,13,46]), 30-day hospital readmission [37] with days in hospital as exposure, bank-marketing subscription, Spambase, and USPS digits (≥ 5 versus < 5), the last as raw pixels and as 10 principal components. Each state is the first round of a delayed replay with 68 historical labels, 14 pending and 24 candidate records; candidates are drawn at random (*natural* stream) or, for credit and readmission, also from an *engineered* stream that ranks covariates into low-risk, pending-aligned, complementary and tail candidates. Controlled worlds and *semi-synthetic* states (real covariates, outcomes from a known logit) provide known p.

Estimated probabilities recover known-probability values. On 512 semi-synthetic states, boosting $\hat{p}$ fitted on auxiliary records labeled from the true p gives estimates whose bias against the known-p protocol lies within ± 0.5 in 95 of 96 checks (8 groups $\times$ 12 quantities); mean biases of H_{post}, H_{pre}, the floor and the main strategy gains are at most 0.20, with state-level correlations of 0.73–0.95 for H_{pre}. A logistic $\hat{p}$ underestimates H_{pre} by 0.26–0.28 where the truth is nonlinear, so the estimator needs a flexible $\hat{p}$.

Recorded-outcome checks and sensitivity. A confidence interval containing zero does not establish agreement. In the revision we use equivalence tests with margin 0.5 in the displayed units. Pooled checks for logistic and boosting learners support equivalence, while only two of 16 individual cohort–learner cells establish it; the neural learner does not establish pooled equivalence. In 192 additional greedy-calibration states, the pooled reported-gap difference has 90% interval $[-0.089, 0.416]$, within the margin, whereas the held-out-gap interval $[-0.138, 0.513]$ crosses it. Known-probability perturbations give mean floor bias about -0.04 at the unperturbed model, but changing probability sharpness shifts bias by approximately 0.5–0.8. Disjoint fitting is essential but does not establish robustness to probability-model misspecification.

Family sensitivity. Enlarging each family with 64 random feasible sets (from 22–27 to 81–88 sets) in these experiments leaves the estimated deployable share $H_{\text{pre}}/H_{\text{post}}$ essentially unchanged: 80–83% in controlled worlds at every subfamily size from 2 to 80, and 6–38% on semi-synthetic arms, where it rises slightly rather than falls with $|\mathcal{F}|$.

6 Findings

The greedy benchmark gap shrinks under held-out selection. Table 1 retains the original cross-fitted greedy estimates. Its columns implement the accounting

Table 1. Benchmark greedy oracle under the risk budget, logistic learner ($\times 10^{-3}$ Brier). Reported gap = test reuse + greedy residual + cross-fitted greedy gain. Last column: lead over the best fixed practical strategy, as reported → after removing reuse and floor.

Setting	States	Reported	Reuse	Residual	CF greedy	Lead over best
Controlled worlds	64	29.5	4.5	3.5	21.5	15.8→7.8
Semi-synthetic (known p)	256	7.2	2.9	3.9	0.3	6.9→0.0
Credit, engineered stream	128	8.0	0.6	4.1	3.3	6.7→2.0
Credit, natural stream	128	3.4	0.8	1.7	0.9	3.2→0.6
Readmission, both streams	64	2.3	0.3	1.4	0.5	2.0→0.3

Table 2. Benchmark oracle along a label-noise gradient (natural stream, 48 states per cohort with the logistic and 32 with the neural learner; $\times 10^{-3}$ Brier). Noise: mean $\hat{p}(1-\hat{p})$. Reuse and floor: % of the reported gap. Lead: over the best fixed strategy, reported → deployable.

		Logistic learner				Neural learner (MLP)			
Cohort	Noise	Gap	Reuse	Residual	Lead	Gap	Reuse	Residual	Lead
USPS, raw pixels	0.036			–		18.7	32	10	15.9→8.0
Spambase	0.054	7.5	32	16	5.1→1.5	14.1	25	29	11.4→3.8
USPS, 10 PCs	0.065	10.8	12	13	7.7→5.0	16.6	23	29	15.0→6.4
Readmission	0.081	1.3	29	66	1.1→−0.1	4.8	23	54	4.0→0.3
Bank Marketing	0.091	4.5	15	39	4.1→1.7	11.9	34	29	10.3→2.8
HMEQ	0.101	3.4	15	42	3.0→1.0	7.8	26	39	7.2→2.1
PPDai	0.108	2.8	17	60	2.7→0.5	10.4	20	38	9.1→3.0
UCI Credit	0.136	3.4	15	67	3.2→0.4	8.2	47	38	7.8→0.8
LendingClub	0.144	2.7	8	79	2.8→0.4	8.5	39	53	8.6→0.7

identity in Section 3; the residual column includes approximation error and is not the fixed-family F. On semi-synthetic states the reported lead over the best fixed strategy is 6.9, while the cross-fitted greedy lead is approximately zero. On recorded-outcome states, correction reduces the apparent lead substantially. This demonstrates sensitivity to evaluation design, without proving that the entire removed component is irreducible.

Noise ordering and a direct intervention. The original cross-cohort study registered a relationship between the noise index and the greedy residual share (Table 2). Its interpretation depends on the ex-ante comparator: using a post-hoc strengthened comparator changes the logistic rank correlation to 0.93 and the neural correlation to 0.52 ($p = 0.15$). The latter no longer supplies clear cross-cohort evidence. A distinct registered intervention flips labels with probability $0, 0.1, 0.2, 0.3$ within HMEQ, Bank Marketing and USPS, using 16 states per dose and cohort. The empirical fixed-family floor increases monotonically in all three cohorts (Figure 1b). This intervention supports a response to randomized label corruption within the specified simulation; it does not establish a universal monotonic relationship for natural label noise.

Bounds, search, and scale checks. The corrected empirical lower and upper inequalities hold in all 624 revision states. Their validity is algebraic; this check

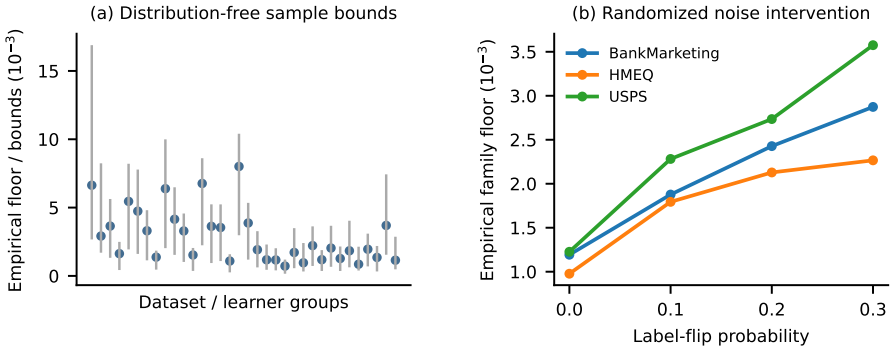


Fig. 1. Corrected diagnostics. (a) Empirical lower bound, floor and upper bound by group, covering 624 states; this verifies the empirical identity, not population coverage. (b) Within-cohort randomized label flips: empirical family floor at four doses, 16 states per dose and cohort. Lines connect descriptive means.

verifies implementation, not a new population confidence guarantee. Stress tests with negative dependence, skewness and heavy tails invalidate the former independent-normal shortcut. In 12-candidate pools, exact enumeration of all feasible sets (about 730 on average) yields similar aggregate greedy diagnostics: the mean residual difference is approximately -0.02 and the ex-ante gain difference 0.01 in displayed units. This checks approximation at that scale. Candidate pools of 24, 48 and 96 produce heterogeneous floor changes, so no large-pool approximation guarantee follows.

Budget sensitivity is empirical. Increasing a budget can enlarge a feasible family, but $F = H_{\text{post}} - H_{\text{pre}}$ is a difference of two changing quantities and need not increase. In the registered budget experiment, varying the target count from 3 to 6 to 12 increased the measured family floor in all eight cohorts, with paired mean increase 0.74 $[0.67, 0.82]$ in displayed units (Figure 2). This is a result for the tested distributions and learner, rather than a consequence of (4).

Who closes the estimated ex-ante selector gain depends on the learner. Table 3 reports, on the same real-label states, the estimated ex-ante selector gain and the gains of three strategies under two learners. With the logistic learner, information proxies derived from that model close a large part of it where it is largest: on engineered HMEQ and LendingClub TargetVar-greedy and Proxy-argmax capture 1.5–2.1 of 3.2–3.4 (in an independent 128-seed batch, 2.5–2.9 of 4.9–5.7, while DC-PRISM trails TargetVar-greedy by 2.8 $[0.0, 5.6]$ on LendingClub). With the boosting learner the estimated ex-ante selector gain is still 0.7–2.7, DC-PRISM's selection gap is material in six of eight credit cells, and no strategy captures more than 0.65: proxies that match the logistic model do not transfer

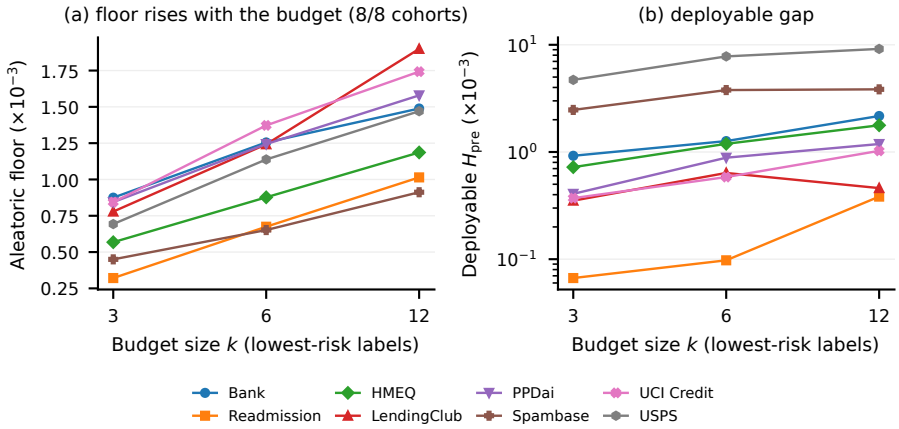


Fig. 2. Risk budget as a lever (logistic learner, 32 states per cohort and budget). The budget is sized for the k lowest-risk candidates. (a) The aleatoric floor rises monotonically in all eight cohorts. (b) The deployable gap (log scale) moves much less uniformly.

to the boosting learner. On readmission the estimated ex-ante selector gain is small (0.7–0.9) and no strategy differs decisively.

DC-PRISM in controlled worlds. In worlds built around directional pending information, 83% of hindsight headroom is estimated as ex-ante value and DC-PRISM captures two thirds of it: its gain over Safe-MinRisk is 12.2 $[7.6, 16.8]$ with 3.1 instead of 6 labels, and remains material in all twelve generator variants we tried (10.8–31.4; pending overlap, risk–information coupling, no misspecified tail, 48 candidates with $K = 12$). A common-family ablation attributes this gain to the proxy evaluated over an anchored family (Proxy-argmax alone gains 13.1 $[6.1, 20.2]$); the conservative layer adds an unconfirmed 3.0 $[-0.2, 6.2]$, and pending-awareness is not isolated by our evidence.

7 Outcome Check: Delayed Replay

The protocol scores single rounds. To check its ordering under dynamics we replayed ten rounds (exploration labels after one round, ordinary labels after two; ALC-Brier averages ten round-start scores) with 512 fresh seeds per credit cell and 256 per readmission stream, numbers registered from the variance of an earlier batch. Table 4 reports a separate dynamic comparison. Where information-greedy rules captured the deployable gap and DC-PRISM did not (engineered HMEQ and LendingClub), DC-PRISM is materially worse than Fisher-cost, BAIT-greedy and TargetVar-greedy (2.5–3.8); where its gain over the quantity reference was significant, it beats Safe-MinRisk and fixed count (2.2–3.1); on engineered readmission it beats BAIT-greedy and TargetVar-greedy (1.3, 1.4); and

Table 3. Cross-fitted finite-family selector gain and gains over M_K on real-label states (64 credit and 128 readmission states per cell; $\times 10^{-3}$ Brier). DC: DC-PRISM; TV: TargetVar-greedy; Proxy: Proxy-argmax.

	Logistic learner				Boosting learner			
State	CF gain	DC	TV	Proxy	CF gain	DC	TV	Proxy
HMEQ, eng.	3.21	0.09	1.51	1.64	2.32	0.48	0.15	0.44
LC, eng.	3.37	1.15	2.01	2.12	1.53	0.05	−0.42	−0.19
PPDai, eng.	1.87	0.37	0.04	0.07	1.18	0.33	0.17	0.11
UCI, eng.	0.16	−0.08	−0.08	−0.08	0.84	0.36	0.42	0.34
HMEQ, nat.	1.26	−0.22	0.21	0.12	2.66	0.24	0.64	0.65
LC, nat.	0.91	−0.04	−0.20	−0.05	1.45	−0.07	0.04	−0.04
PPDai, nat.	0.59	0.10	−0.17	0.02	1.17	−0.03	−0.18	−0.10
UCI, nat.	0.92	0.07	0.19	0.32	1.60	0.03	0.13	0.08
Readm., eng.	0.84	0.09	−0.14	−0.34	0.92	−0.01	−0.25	−0.28
Readm., nat.	0.67	0.06	0.30	0.36	0.72	0.00	−0.10	−0.07

Table 4. Delayed replay: comparator minus DC-PRISM ALC-Brier ($\times 10^{-3}$; positive favors DC-PRISM; bold: material under the registered family).

	Credit, engineered				Credit, natural				Readmission	
Comparator	HMEQ	LC	PPDai	UCI	HMEQ	LC	PPDai	UCI	eng.	nat.
Safe-MinRisk	**2.4**	0.4	**3.1**	−0.1	0.2	0.2	0.4	0.1	0.0	0.2
Fixed count	**2.2**	0.4	**2.4**	−0.1	0.1	0.1	0.3	0.0	0.4	0.2
Random	0.6	−0.3	**2.4**	0.0	0.3	0.5	0.6	0.4	0.8	0.5
Uncertainty	0.5	1.4	0.7	0.0	0.4	0.4	0.1	0.1	0.4	0.0
Fisher-cost	**−3.8**	**−3.3**	0.8	−0.1	−0.3	−0.1	0.1	−0.1	1.0	0.0
BAIT-greedy	**−3.6**	**−2.7**	0.3	0.1	−0.2	0.0	0.1	0.0	**1.3**	0.1
TargetVar-greedy	**−3.5**	**−2.5**	0.0	0.0	−0.2	−0.1	0.0	−0.1	**1.4**	0.0

where little deployable gap exists (natural streams, engineered UCI) no comparison is material. Realized adverse exposure exceeded the envelope in 0–4 of 40,960 acquisition rounds per credit policy and in none of 5,120 for readmission.

8 Related Work

Evaluating active learning. Recent work shows that active-learning gains are fragile under realistic evaluation [28], sensitive to uncontrolled randomness [19] and inconsistent across domains [42,43], and uses oracles to bound attainable performance [43,18]. Werner et al. note that their greedy oracle overfits the test set [42]. We quantify that reuse and add an aleatoric component that remains after it is removed.

Regret notions and uncertainty. Ex-ante and ex-post regret are distinguished in decision-focused learning and robust optimization [34]; expected regret exceeds pseudo-regret by a noise term in bandits [6,2]; proper scoring rules such as the Brier score separate aleatoric from epistemic uncertainty [14]; and selecting the best of noisy estimates is optimistic [36,7], the value of perfect information being classical [16,31]. We bring these ideas to oracle bounds for label acquisition and make the aleatoric part measurable and predictable on real data.

Acquisition under risk and delay. Acquisition criteria range from uncertainty and transductive design to Fisher information, mutual and predictive information and decision loss [10,35,27,47,1,23,3,17,8]; budgets and knapsack constraints are handled by greedy selection [24,39,40]. Selective labels [25,41,22,21,11], safe exploration and safe active learning [38,33,45] and delayed feedback [20,9,30] motivate the risk-constrained setting in which we study the floor.

9 Discussion and Conclusion

Oracle gaps depend on what the selector may observe and how its performance is evaluated. For a fixed family, the value of observing hidden training outcomes is distinct from ex-ante acquisition value. Greedy benchmark comparisons additionally mix evaluation-set reuse, search error and finite-simulation selection. The corrected inequalities, noise interventions and approximation checks support an evaluation audit with explicit assumptions. A useful report pairs held-out oracle performance with fixed-family diagnostics and sensitivity to the probability model. DC-PRISM's mixed replay performance illustrates why substantial hindsight opportunity alone cannot establish the effectiveness of an acquisition strategy.

Limitations. Probability-model substitution does not identify the real conditional outcome law, and test-label conditioning is weaker than population predictive-risk evaluation. Cross-fitted greedy performance includes optimization and Monte Carlo error; post-hoc maxima over policy estimates are optimistic. Exact enumeration covers 12-candidate pools only. Random label corruption is a controlled intervention rather than a complete model of natural uncertainty. Pools and learners remain modest; larger pretrained models and verified calendar-time cohorts were not studied. Baselines are explicit adaptations. The risk-envelope probability interpretation additionally requires conditional independence and valid upper risk scores, which the empirical scores do not establish. Public source and local results have different coverage; anonymous artifact access must be prepared and tested before submission.

References

1. Ash, J., Goel, S., Krishnamurthy, A., Kakade, S.: Gone fishing: Neural active learning with Fisher embeddings. In: Advances in Neural Information Processing Systems. vol. 34, pp. 8927–8939 (2021), https://proceedings.neurips.cc/paper_files/paper/2021/hash/4afe044911ed2c247005912512ace23b-Abstract.html
2. Audibert, J.Y., Bubeck, S.: Regret bounds and minimax policies under partial monitoring. Journal of Machine Learning Research **11**, 2785–2836 (2010)
3. Bickford Smith, F., Kirsch, A., Farquhar, S., Gal, Y., Foster, A., Rainforth, T.: Prediction-oriented Bayesian active learning. In: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 206, pp. 7331–7348 (2023), https://proceedings.mlr.press/v206/bickfordsmith23a.html

4. Boucheron, S., Lugosi, G., Massart, P.: Concentration Inequalities: A Nonasymptotic Theory of Independence. Oxford University Press (2013). https://doi.org/10.1093/acprof:oso/9780199535255.001.0001
5. Brier, G.W.: Verification of forecasts expressed in terms of probability. Monthly Weather Review **78**(1), 1–3 (1950). https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2
6. Bubeck, S., Cesa-Bianchi, N.: Regret analysis of stochastic and nonstochastic multi-armed bandit problems. Foundations and Trends in Machine Learning **5**(1), 1–122 (2012)
7. Cawley, G.C., Talbot, N.L.C.: On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research **11**, 2079–2107 (2010)
8. Chaloner, K., Verdinelli, I.: Bayesian experimental design: A review. Statistical Science **10**(3), 273–304 (1995)
9. Chapelle, O.: Modeling delayed feedback in display advertising. In: Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 1097–1105 (2014)
10. Cohn, D.A., Ghahramani, Z., Jordan, M.I.: Active learning with statistical models. Journal of Artificial Intelligence Research **4**, 129–145 (1996). https://doi.org/10.1613/jair.295
11. Crook, J., Banasik, J.: Does reject inference really improve the performance of application scoring models? Journal of Banking & Finance **28**(4), 857–874 (2004)
12. He, H.: PPDaiData. figshare dataset, CC BY 4.0 (2021). https://doi.org/10.6084/m9.figshare.13573856.v1, 55,596 loans from the PaiPaiDai P2P lending platform
13. He, H., Zhang, W., Zhang, S.: A novel ensemble method for credit scoring: Adaption of different imbalance ratios. Expert Systems with Applications **98**, 105–117 (2018). https://doi.org/10.1016/j.eswa.2018.01.012
14. Hofman, P., Sale, Y., Hüllermeier, E.: Quantifying aleatoric and epistemic uncertainty with proper scoring rules. arXiv:2404.12215 (2024)
15. Holm, S.: A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics **6**, 65–70 (1979). https://doi.org/10.2307/4615733
16. Howard, R.A.: Information value theory. IEEE Transactions on Systems Science and Cybernetics **2**(1), 22–26 (1966)
17. Huang, Z., Bickford Smith, F., Rainforth, T.: Loss-driven Bayesian active learning. In: Proceedings of the 29th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 300, pp. 5140–5148 (2026), https://proceedings.mlr.press/v300/huang26b.html
18. Huseljic, D., Hahn, P., Herde, M., Sandrock, C., Sick, B.: BoSS: A best-of-strategies selector as an oracle for deep active learning. Transactions on Machine Learning Research (2026), arXiv:2603.13109
19. Ji, Y., Kaestner, D., Wirth, O., Wressnegger, C.: Randomness is the root of all evil: More reliable evaluation of deep active learning. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). pp. 3943–3952 (2023)
20. Joulani, P., György, A., Szepesvári, C.: Online learning under delayed feedback. In: Proceedings of the 30th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 28, pp. 1453–1461 (2013)
21. Keswani, V., Mehrotra, A., Celis, L.E.: Fair classification with partial feedback: An exploration-based data collection approach. In: Proceedings of the 41st Inter-

national Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 235, pp. 23547–23576 (2024), https://proceedings.mlr.press/v235/keswani24a.html
22. Kilbertus, N., Gomez Rodriguez, M., Schölkopf, B., Muandet, K., Valera, I.: Fair decisions despite imperfect predictions. In: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 108, pp. 277–287 (2020)
23. Kirsch, A., van Amersfoort, J., Gal, Y.: BatchBALD: Efficient and diverse batch acquisition for deep Bayesian active learning. arXiv:1906.08158 (2019). https://doi.org/10.48550/arXiv.1906.08158
24. Krause, A., Guestrin, C.: Near-optimal nonmyopic value of information in graphical models. In: Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI). pp. 324–331 (2005)
25. Lakkaraju, H., Kleinberg, J., Leskovec, J., Ludwig, J., Mullainathan, S.: The selective labels problem: Evaluating algorithmic predictions in the presence of unobservables. In: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 275–284 (2017). https://doi.org/10.1145/3097983.3098066
26. LendingClub Corporation: LendingClub historical loan data (accepted loans), originally published by LendingClub; processed copy obtained from the DePaul University Economics Data Archive, https://bigblue.depaul.edu/jlee141/econdata/LendingClub_LoanData/, accessed 4 November 2026
27. Lewis, D.D., Gale, W.A.: A sequential algorithm for training text classifiers. In: Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 3–12 (1994)
28. Lüth, C.T., Bungert, T.J., Klein, L., Jaeger, P.F.: Navigating the pitfalls of active learning evaluation: A systematic framework for meaningful performance assessment. In: Advances in Neural Information Processing Systems. vol. 36, pp. 9789–9836 (2023)
29. MacKay, D.J.C.: A practical Bayesian framework for backpropagation networks. Neural Computation **4**(3), 448–472 (1992). https://doi.org/10.1162/neco.1992.4.3.448
30. Pham, T., Kottke, D., Krempl, G., Sick, B.: Stream-based active learning for sliding windows under the influence of verification latency. Machine Learning **111**(6), 2011–2036 (2022). https://doi.org/10.1007/s10994-021-06099-z
31. Raiffa, H., Schlaifer, R.: Applied Statistical Decision Theory. Harvard University Press, Boston (1961)
32. SAS Institute: Home equity data set. SAS example datasets, https://support.sas.com/documentation/onlinedoc/viya/exampledatasets/home_equity.csv, accessed 4 November 2026
33. Schreiter, J., Nguyen-Tuong, D., Eberts, M., Bischoff, B., Markert, H., Toussaint, M.: Safe exploration for active learning with Gaussian processes. In: Machine Learning and Knowledge Discovery in Databases (ECML PKDD). LNCS, vol. 9286, pp. 133–149. Springer (2015)
34. Schutte, N., Postek, K., Yorke-Smith, N.: Robust losses for decision-focused learning. arXiv:2310.04328 (2023)
35. Settles, B.: Active learning literature survey. Computer Sciences Technical Report 1648, University of Wisconsin–Madison (2009)
36. Smith, J.E., Winkler, R.L.: The optimizer's curse: Skepticism and postdecision surprise in decision analysis. Management Science **52**(3), 311–322 (2006)

37. Strack, B., DeShazo, J.P., Gennings, C., Olmo, J.L., Ventura, S., Cios, K.J., Clore, J.N.: Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. BioMed Research International **2014**, 781670 (2014). https://doi.org/10.1155/2014/781670
38. Sui, Y., Gotovos, A., Burdick, J.W., Krause, A.: Safe exploration for optimization with Gaussian processes. In: Proceedings of the 32nd International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 37, pp. 997–1005 (2015)
39. Sviridenko, M.: A note on maximizing a submodular set function subject to a knapsack constraint. Operations Research Letters **32**(1), 41–43 (2004)
40. Tomanek, K., Hahn, U.: A comparison of models for cost-sensitive active learning. In: Coling 2010: Posters. pp. 1247–1255 (2010), https://aclanthology.org/C10-2143/
41. Wei, D.: Decision-making under selective labels: Optimal finite-domain policies and beyond. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 11035–11046 (2021), https://proceedings.mlr.press/v139/wei21a.html
42. Werner, T., Burchert, J., Schmidt-Thieme, L.: Towards comparable active learning. arXiv:2311.18356 (2023)
43. Werner, T., Burchert, J., Stubbemann, M., Schmidt-Thieme, L.: A cross-domain benchmark for active learning. arXiv:2408.00426 (2024)
44. Wilson, E.B.: Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association **22**(158), 209–212 (1927). https://doi.org/10.1080/01621459.1927.10502953
45. Wu, Y., Shariff, R., Lattimore, T., Szepesvári, C.: Conservative bandits. In: Proceedings of the 33rd International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 48, pp. 1254–1262 (2016)
46. Yeh, I.C., Lien, C.h.: The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. Expert Systems with Applications **36**(2), 2473–2480 (2009). https://doi.org/10.1016/j.eswa.2007.12.020
47. Yu, K., Bi, J., Tresp, V.: Active learning via transductive experimental design. In: Proceedings of the 23rd International Conference on Machine Learning. pp. 1081–1088 (2006). https://doi.org/10.1145/1143844.1143980